NISHANT SINGH

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PROFESSIONAL SUMMARY

AI / Machine Learning Engineer with experience building end-to-end ML systems including **data** preprocessing model Training **evaluation, and scalable deployment**. Skilled in **Python, PyTorch, TensorFlow and LLM-based Application** Such as RAG and embedding.

EDUCATION

Master of Computer Applications (MCA) — Pimpri Chinchwad University, Pune (*Aug 2024 – Jun 2026)*

Bachelor of Computer Applications (BCA) — Mahatma Gandhi Kashi Vidyapith (MGKVP), Varanasi *(Sep 2020 – Sep 2023)*

SKILLS

Programming & Data: Python, SQL
Libraries: Pandas, NumPy, Matplotlib, Seaborn
Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, MLAlgo, Feature Engineering, Model Training, Model Evaluation, Cross-Validation, Hyperparameter Tuning
Deep Learning / NLP: Transformers, LLM Applications, Prompt Engineering, RAG, Embeddings
Data Engineering: Data Cleaning, Data Pipelines, ETL, Feature Selection
Computer Vision: OpenCV, YOLO
Backend & APIs: Fast API
MLOps & Tools: MLflow, Docker, CI/CD, Git, GitHub
Cloud & Deployment: AWS (EC2, S3), Flask, Streamlit

PROJECTS

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- **LLM-Integrated Learning Assistant — Gemini API, FastAPI, Retrieval** *(Aug 2025 – Sept 2025)*
  • Implemented Q&A/explain/quiz flows grounded to curated notes with **content-safety checks**.
  • Exposed REST endpoints; added **rate-limit guards** and basic usage analytics.
  • Improved **user thumbs-up rate by ~10–12%** via prompt and context-window tuning
- **Real-Time Object Detection & Alert System — YOLO, OpenCV, FastAPI, WebSockets** *(Mar 2025 – Apr 2025)*
  • Built YOLOv4 + OpenCV microservice behind FastAPI; streamed detections via WebSockets.
  • Optimized preprocessing/IO to cut **alert latency from ~1.2s to <400ms** on commodity hardware.
  • Reduced **false positives by ~15–20%** using confidence calibration + rule-based filters.
- **Hybrid Music Recommendation — TF-IDF, Fuzzy Matching, Cosine Similarity** *(Feb 2025 – Mar 2025)*
  • Engineered lyrics/genre features and metadata embeddings for candidate generation and ranking.
  • Added re-ranker + deduplication to boost **top 10 hit-rate by ~10–12%**.
  • Published a **clean API contract** for UI integration.
- **Housing Price Prediction Pipeline — ETL, Feature Engineering, Regression** *(Jan 2025– Feb 2025)*
  • Built reproducible ETL: imputation, outlier handling, scaling, feature selection.
  • Benchmarked Ridge/Lasso/Random Forest; reported **MAE/MAPE with K-fold CV**.
  • Tracked experiments in **MLflow**; packaged the model for REST inference.

CERTIFICATIONS

- DeepLearning.AI — Machine Learning Specialization, via Coursera, 2025
- Microsoft — Azure Machine Learning, via Coursera, 2025
- Udemy — Data Science & ML Bootcamp, 2025

CO-CURRICULAR

- Coordinated technical workshops and college IT mela
- Winner/placed in inter-college quizzes (SMS, Kashi Vidyapith, 2023).